Problem Set 4F

Geometric proofs

Do these after the slides for Section 4F. Textbook Exercise 4F. Answers are on the last page. Show full working in your exercise book.

Skills

1 Prove that the midpoints of the sides of any quadrilateral are the vertices of a parallelogram.

2 Prove that the diagonals of a rhombus are perpendicular.

3 Prove that if the diagonals of a parallelogram are perpendicular, then it is a rhombus.

4 Prove that the diagonals of a rectangle are equal in length.

5 Triangle OAB has a right angle at O, and M is the midpoint of AB. Prove that $OM = \frac{1}{2}AB$.

Exam style

6 In triangle OAB, M is the midpoint of AB and N is the midpoint of OB. G lies on OM with $\overrightarrow{OG} = \frac{2}{3}\overrightarrow{OM}$. Prove that G lies on AN.

7 Use $\overrightarrow{AB} = \mathbf{b} - \mathbf{a}$ to prove the cosine rule for triangle OAB.

8 In triangle ABC, the altitudes from A and B meet at H. Taking H as the origin, prove that CH is perpendicular to AB.

Problem Set 4F: answers

1 With midpoints P, Q, R, S of AB, BC, CD, DA: $\overrightarrow{PQ} = \frac{1}{2}(\mathbf{c} - \mathbf{a}) = \overrightarrow{SR}$, so $PQRS$ is a parallelogram

2 $(\mathbf{a} + \mathbf{c}) \cdot (\mathbf{c} - \mathbf{a}) = |\mathbf{c}|^2 - |\mathbf{a}|^2 = 0$ since $|\mathbf{a}| = |\mathbf{c}|$

3 $(\mathbf{a} + \mathbf{c}) \cdot (\mathbf{c} - \mathbf{a}) = 0$ gives $|\mathbf{c}|^2 = |\mathbf{a}|^2$, so adjacent sides are equal

4 $\mathbf{a} \cdot \mathbf{c} = 0$, so $|\mathbf{a} + \mathbf{c}|^2 = |\mathbf{a}|^2 + |\mathbf{c}|^2 = |\mathbf{c} - \mathbf{a}|^2$

5 $|\overrightarrow{OM}|^2 = \frac{1}{4}|\mathbf{a} + \mathbf{b}|^2 = \frac{1}{4}(|\mathbf{a}|^2 + |\mathbf{b}|^2) = \frac{1}{4}|\mathbf{b} - \mathbf{a}|^2$ since $\mathbf{a} \cdot \mathbf{b} = 0$

6 $\overrightarrow{AG} = \frac{1}{3}\mathbf{b} - \frac{2}{3}\mathbf{a} = \frac{2}{3}\overrightarrow{AN}$, so A, G, N are collinear

7 $|\overrightarrow{AB}|^2 = (\mathbf{b} - \mathbf{a}) \cdot (\mathbf{b} - \mathbf{a}) = |\mathbf{a}|^2 + |\mathbf{b}|^2 - 2|\mathbf{a}||\mathbf{b}| \cos\theta$

8 $\mathbf{a} \cdot (\mathbf{c} - \mathbf{b}) = 0$ and $\mathbf{b} \cdot (\mathbf{c} - \mathbf{a}) = 0$ give $\mathbf{a} \cdot \mathbf{c} = \mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{c}$, so $\mathbf{c} \cdot (\mathbf{b} - \mathbf{a}) = 0$